Exploring the Impact of Financial News Sentiment on Stock Price Forecasting: A Comparative Deep Learning Approach

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***Abstract*—The application of Artificial Intelligence (AI) in stock price prediction has demonstrated significant advancements, with Machine Learning and Deep Learning techniques proving highly efficient in this domain. Two widely adopted architectures for stock price prediction are the Long Short Term Memory (LSTM) and Gated Recurrent Unit (GRU) models. Recognizing the potential impact of financial news sentiment on forecasting, this study investigates whether incorporating such sentiment yields superior results compared to relying solely on historical stock prices. Accurate predictions in the financial market are challenging due to its inherent high volatility and non-linear nature. To contribute insights into the effectiveness of various deep learning architectures, a comparative analysis was conducted on Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Convolutional Neural Network (CNN), and the Prophet model. The evaluation focused on three companies listed on the National Stock Exchange (NSE). The study employs the Mean Squared Error (MSE) metric to rigorously assess and compare the models' performance. The results of the comparative analysis provide valuable implications for precision in stock price prediction within the context of market volatility and non-linearity. The study adds to the growing body of research on the application of AI in financial forecasting, emphasizing the importance of considering sentiment analysis alongside traditional approaches for enhanced predictive accuracy.**

***Index Terms*—LSTM, GRU, Prophet, Time Series Analysis, National Stock Exchange (NSE)**

I. Introduction

Stock price prediction is crucial in finance. Time series forecasting, used extensively, tracks stock price dynamics. This research aims to improve predictive accuracy. Stock price prediction is challenging due to volatility and unpredictability, making enhancement crucial for better financial analysis.

This forecasting challenge can be categorized into three distinct timeframes:

a. Short-term forecasting, encompasses predictions over brief periods such as seconds, minutes, hours, few days, weeks, or months.
b. Medium-term forecasting, extending the prediction horizon to one to two years.
c. Long-term forecasting, involving predictions over periods exceeding two years.

Numerous models and techniques have been developed and utilized for stock price prediction. Our focus lies within time series forecasting, broadly categorized into two classes:

a. *Linear Models:* This encompasses ARIMA [1] and its variations, including SARIMA [2]. These models utilize predefined equations to fit mathematical models to univariate time series data. Our paper will delve into the ARIMA model, discussed in a subsequent section.
b. *Non-Linear Models:* This category includes deep learning algorithms, GARCH [3], and others. Deep learning algorithms, renowned for capturing non-linear patterns, are of particular interest. In our research, we employ Convolutional Neural Network (CNN) [4] and Long Short-Term Memory (LSTM) models. LSTMs, integral to Recurrent Neural Networks (RNNs), possess the ability to retain input information over extended periods, a crucial aspect in stock price prediction for enhanced precision and accuracy.

Given the vast and highly non-linear nature of stock market data, the preference for deep learning models is evident. However, working with time series data demands careful consideration of certain aspects:

a. *Stationarity:* This involves assessing statistical properties such as mean, variance, covariance, and standard deviation, ensuring they remain constant over time for a stationary time series.
b. *Seasonality:* Recognizing periodic fluctuations or patterns within the time series, denoted as seasonality, is crucial.

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Any predictable oscillation repeating over time falls within this category.

c. *Autocorrelation:* This metric measures the correlation between a variable's current value and its past values, indicating the degree of correlation across different successive time intervals.

In summary, the immense and non-linear nature of stock market data necessitates sophisticated models, and our research aims to contribute to this domain by exploring and enhancing the performance of both linear and non-linear time series forecasting models.

The paper's structure includes: Section 2 reviews related work. Section 3 details research methodology. Section 4 discusses results. Section 5 concludes, summarizing findings and exploring future study scope.

II. Related Background

Recent literature explores machine-learning and deep-learning techniques [6] for stock price prediction, demonstrating their efficacy across diverse datasets and methods.

In a notable study [7], the Autoregressive Integrated Moving Average (ARIMA) model [5] was utilized for stock price prediction. The research titled "Stock Price Prediction Using the ARIMA Model" detailed the construction of predictive models using data from the Nigeria Stock Exchange (NSE) and New York Stock Exchange (NYSE), highlighting the model's potential for short-term predictions.

In [9], the same researchers explored stock price prediction using Artificial Neural Networks (ANN) [8], introducing a hybridized approach combining technical and fundamental analysis variables. Another study [10] focused on deep learning models such as Long Short-Term Memory (LSTM), Recurrent Neural Network (RNN), and Convolutional Neural Network (CNN) for stock price prediction, emphasizing individual company forecasts and stock index movements using daily closing prices.

In [11], Seasonal Autoregressive Integrated Moving Average (SARIMA) and Prophet Prediction Model were employed for stock price prediction, achieving a root mean square error (RMSE) difference of 44.9545.

These investigations underscore the potential effectiveness of machine learning and deep learning in forecasting stock prices. Applying these methods to predict individual company prices or stock index movements using daily closing prices shows promise for enhancing prediction accuracy. Certain models, such as the Hybrid LSTM-GRU and RNN with LSTM, demonstrate notable efficacy. CNN and ARIMA models perform well in short-term prediction scenarios. However, model performance may vary based on dataset characteristics.

III. Proposed Flow

Figure 1 illustrates the proposed flow of the research.

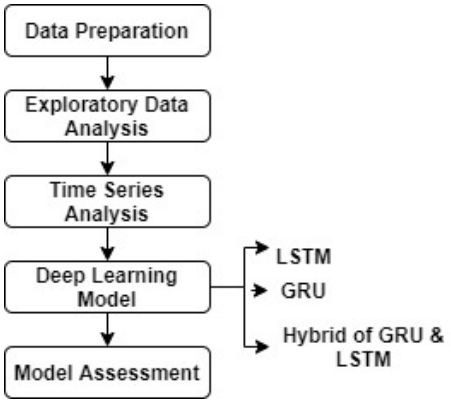

Fig. 1. Proposed Flow

A. Data Preparation

Historical stock price datasets for TCS, ICICI, and Powergrid were procured from the National Stock Exchange (NSE) website, spanning a one-year timeframe. These datasets encompass diverse attributes, including but not limited to 'OPEN', 'Date', 'LOW', 'HIGH,', 'Series, 'PREV. CLOSE', 'Close' 'VWAP','LTP','52W L','52W H','VALUE', 'VOLUME' and '# Trades.' These attributes collectively offer a comprehensive overview of the daily trading activities and performance metrics associated with the respective stocks.

Each dataset (TCS, ICICI, and Powergrid) underwent meticulous analysis using five machine learning models: Long Short-Term Memory (LSTM), Convolutional Neural Network (CNN), AutoRegressive Integrated Moving Average (ARIMA), a Hybrid Model integrating LSTM and Gated Recurrent Unit (GRU), and Facebook's Prophet. This approach systematically explored and evaluated the predictive capabilities of each model, enhancing understanding of their performance on the specific stock datasets.

B. Exploratory Data Analysis

The target variable selected for forecasting is 'close,' with its preceding timestamp being utilized for analysis. Consequently, all the analyses outlined below will be executed with a specific emphasis on this feature.

Rolling Mean and Rolling Std Analysis

The analyses involve the computation of the moving average (Rolling Mean) and the moving standard deviation (Rolling Std) of a time series. These techniques are employed to mitigate short-term fluctuations, accentuate trends, and pinpoint potential periods of volatility. Often utilized as a pre-processing step, they contribute to a more profound comprehension of the underlying patterns in the data. Figure 2 demonstrates the rolling mean and rolling std analysis for Airtel stock prices.

C. Time Series Analysis

i. **Seasonal Decomposition Analysis:** Seasonal decomposition entails the dissection of a time series into its constituent components: trend, seasonality, and residuals. The objective of this analysis is to discern recurring patterns and trends within the data. This step is deemed

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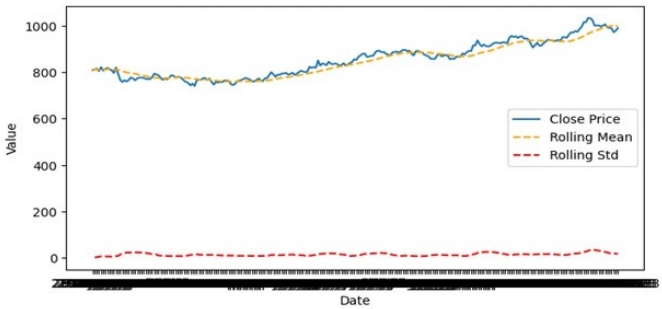

Fig. 2. Airtel Stock Price with Rolling Mean & Rolling Std

crucial in comprehending the cyclic nature inherent in the time series. Figure 3 illustrates the seasonal decomposition analysis of the Powergrid dataset.

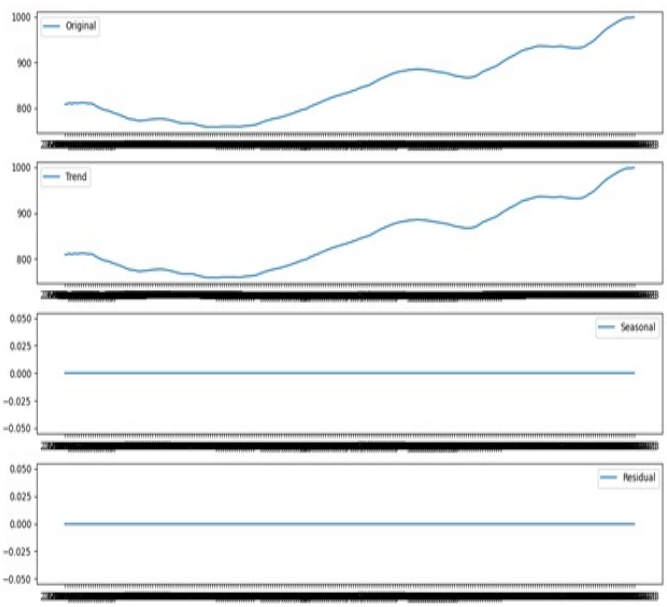

Fig. 3. Seasonal Decomposition Analysis - Powergrid dataset

ii. **ADF Test (Augmented Dickey-Fuller Test):** The ADF test serves as a statistical tool for evaluating the stationarity of a time series, aiding in the determination of whether differencing is requisite to achieve stationarity. Stationarity, a critical assumption for various time series models, is assessed through this test. In this instance, as the p-value exceeds the 5% significance level, the null hypothesis positing data stationarity is rejected. Figure 4 illustrates the Augmented Dickey-Fuller Test on Powergrid dataset.

iii. **ACF (Autocorrelation Function) and PACF (Partial Autocorrelation:** ACF and PACF plots are graphical tools employed for the analysis of autocorrelation and partial autocorrelation in a time series, respectively. The utilization of these plots aids in the identification of patterns and dependencies among observations at various lags.

```
ADF:  -0.15536855235790284
P-Value:  0.9436366650804409
Num Of Lags:  0
Num of observation used for ADF Regression and Critical Values Calculation:  247
Critical Value
        1% :  -3.457105309726321
        5% :  -2.873313676101283
        10% :  -2.5730443824681606
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Fig. 4. ADF Test - Powergrid dataset

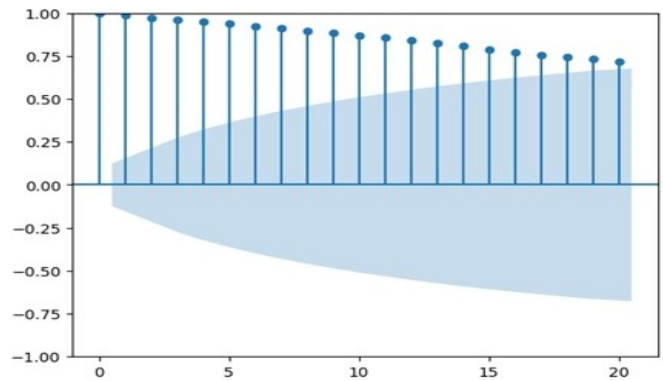

Fig. 5. Autocorrelation - Powergrid dataset

D. Deep-Learning Models

i. **LSTM:** Long Short-Term Memory (LSTM) networks, classified as a type of recurrent neural network (RNN), are well-suited for modeling sequences and time-dependent data. These networks are characterized by memory cells capable of storing and retrieving information over extended sequences. An LSTM cell comprises key components, namely the input gate, forget gate, cell state, and output gate.
The cell state, represented as D_t, undergoes updates based on both the input data and the preceding cell state. The control of this update process is governed by the input gate (j_t), determining the extent to which new information is incorporated into the cell state. The

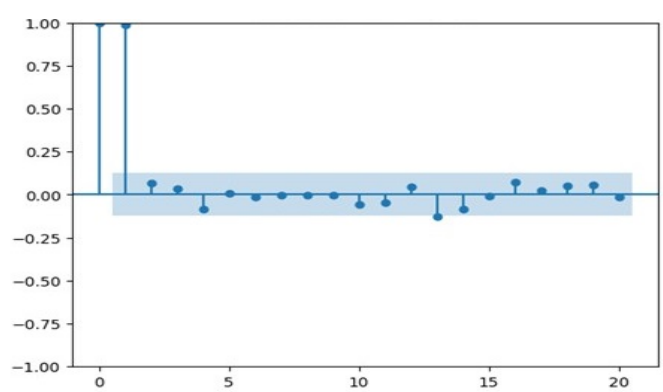

Fig. 6. Partial Autocorrelation - Powergrid dataset

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mathematical representation for cell state and input gate is discussed via Equation 1.

$$\begin{aligned} D_t &= g_t \odot D_{t-1} + j_t \odot \tilde{D}_t \\ \tilde{D}_t &= tanh(X_c \cdot [z_{t-1}, y_t] + a_c) \\ j_t &= \sigma(X_i \cdot [z_{t-1}, y_t] + a_i) \end{aligned} \tag{1}$$

Here, g_t is the forget gate output, $\odot$ denotes element-wise multiplication, σ is the sigmoid activation function, and X_c, X_i are weight matrices.
The output of the LSTM cell, z_t, is determined by the output gate o_t and the updated cell state D_t. The mathematical representation for computing the output of LSTM cell is discussed via Equation 2.

$$\begin{aligned} o_t &= \sigma(W_o \cdot [z_{t-1}, y_t] + a_o) \\ z_t &= o_t \odot tanh(D_t) \end{aligned} \tag{2}$$

Here, W_o is another weight matrix, and σ is the sigmoid activation function.

ii. **GRU:** The Gated Recurrent Unit (GRU), classified as another variant of recurrent neural network (RNN), features a cell with two primary gates: the update gate (z_t) and the reset gate (r_t). These gates govern the information flow within the GRU, facilitating the decision-making process regarding which information to update and what to discard.
The update and reset gates collectively contribute to the modification of the hidden state of the GRU, denoted as z_t. The mathematical representation of GRU is discussed via Equation 3.

$$\begin{aligned} z_t &= \sigma(X_z \cdot [z_{t-1}, y_t] + a_z) \\ r_t &= \sigma(X_r \cdot [z_{t-1}, y_t] + a_r) \\ \tilde{z_t} &= tanh(X_h \cdot [r_t \odot z_{t-1}, y_t] + a_h) \\ z_t &= (1 - z_t) \odot z_{t-1} + z_t \odot \tilde{z_t} \end{aligned} \tag{3}$$

Here, σ is the sigmoid activation function, $\odot$ denotes element-wise multiplication, and X_z, X_r, X_h are weight matrices.

iii. **Hybrid Model of GRU and LSTM:** The hybrid model, integrating both Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), capitalizes on the respective strengths of each architecture to enhance predictive performance. Formulated as a weighted combination of individual LSTM and GRU predictions, the hybrid prediction incorporates an optimized weight parameter, denoted as σ, to achieve optimal predictive accuracy. Equation 4 demonstrates the mathematical representation of the hybrid prediction model.

$$Hybrid = \sigma \times LSTM + (1 - \sigma) \times GRU \tag{4}$$

The parameter σ represents a tunable parameter governing the contribution of each model to the final prediction. The identification of the optimal value for σ involves an iterative process. A for loop is employed to systematically vary σ across a specified range. During each iteration, the mean squared error (MSE) score is computed by comparing the hybrid predictions against the actual values. The σ value corresponding to the minimum MSE score is then chosen as the optimal weighting factor.

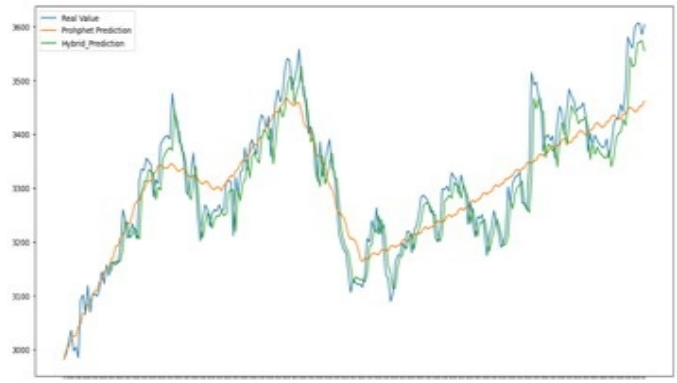

Fig. 7. LSTM-GRU & PROPHET - TCS data

E. Model Assessment

The Mean Squared Error (MSE) stands as a widely utilized metric for evaluating prediction accuracy. It quantifies the average squared difference between predicted and actual values, offering a quantitative assessment of the prediction error. The formulation of MSE is expressed via equation 5.

$$MSE = \frac{1}{N} \sum_{d=1}^{N} (a_d - p_d)^2 \tag{5}$$

In this expression, 'N' signifies the total number of data points, a_d denotes the actual value on day 'd', and p_d represents the corresponding predicted value. The MSE is computed by squaring the difference between each actual and predicted value, summing these squared differences, and subsequently dividing by the total number of data points.

IV. Results & Discussion

This section provides a concise and lucid overview of the results, accompanied by an analysis of their significance within the context of the study topic. The historical prices for TCS, ICICI Bank, and Powergrid spanned a one-year duration and were directly sourced from the National Stock Exchange (NSE). In this study, the training of our model exclusively utilized the 'close' column.

A. TCS

The implementation of the LSTM, LSTM-GRU hybrid model, CNN, and Prophet model was carried out on the TCS dataset. The prediction graphs for the aforementioned models are illustrated via Figure 7, 8 and 9.

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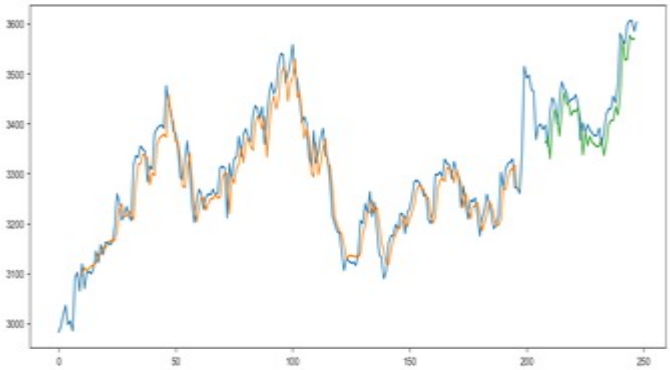

Fig. 8. LSTM model - TCS data

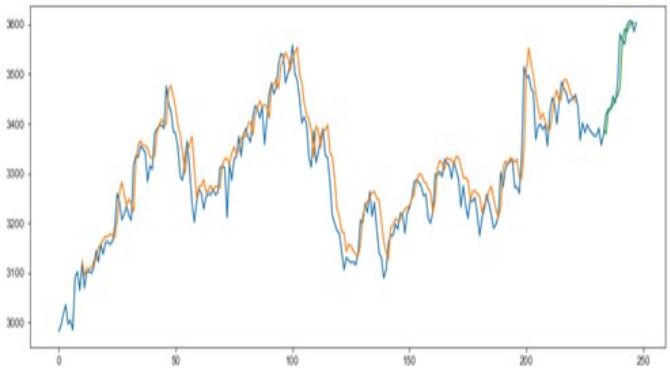

Fig. 9. CNN model - TCS data

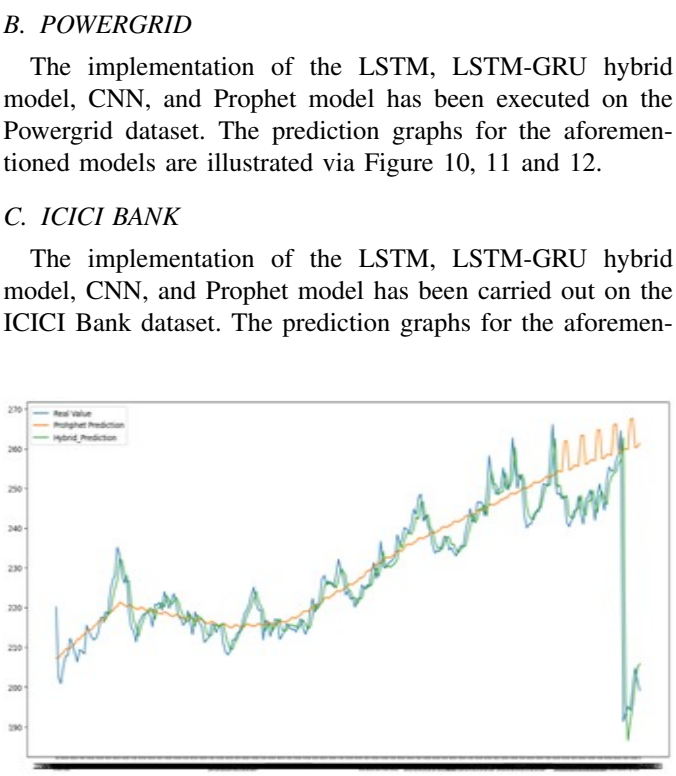

Fig. 10. LSTM-GRU & PROPHET - Powergrid data

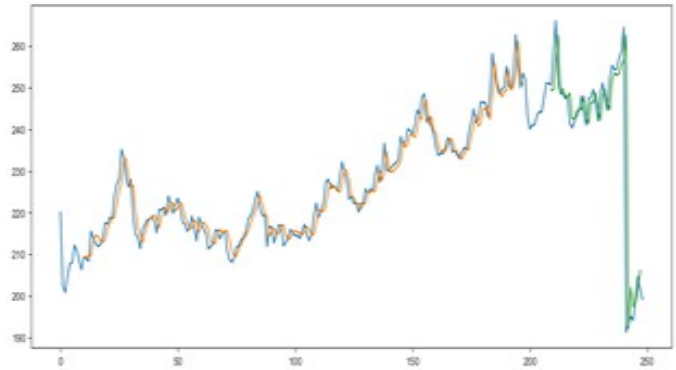

Fig. 11. LSTM model - Powergrid data

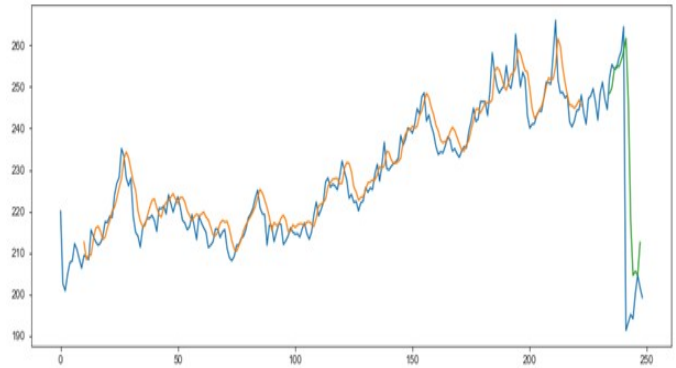

Fig. 12. CNN model - Powergrid data

B. POWERGRID

The implementation of the LSTM, LSTM-GRU hybrid model, CNN, and Prophet model has been executed on the Powergrid dataset. The prediction graphs for the aforementioned models are illustrated via Figure 10, 11 and 12.

C. ICICI BANK

The implementation of the LSTM, LSTM-GRU hybrid model, CNN, and Prophet model has been carried out on the ICICI Bank dataset. The prediction graphs for the aforementioned models are illustrated via Figure 13, 14 and 15. In Figure 7, 10 and 13 green line indicates the 'Hybrid' and the orange indicates the 'Prophet' model. In Figure 8, 11 and 14 orange line indicates training prediction and the green line indicates test prediction.

The calculation of the Root Mean Squared Error (RMSE) score for the aforementioned models has also been undertaken. It can be asserted that, in terms of the RMSE score, the LSTM-GRU Hybrid model demonstrates superior performance in forecasting the TCS stock price.

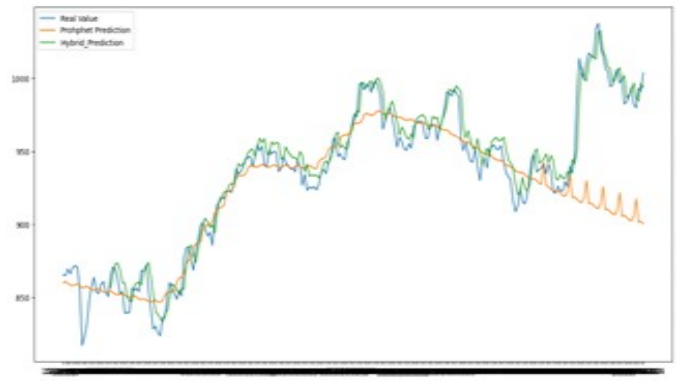

Fig. 13. LSTM-GRU & PROPHET - ICICI BANK data

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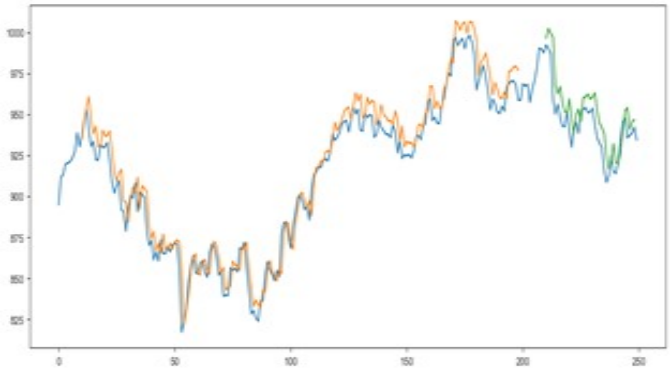

Fig. 14. LSTM model - ICICI BANK data

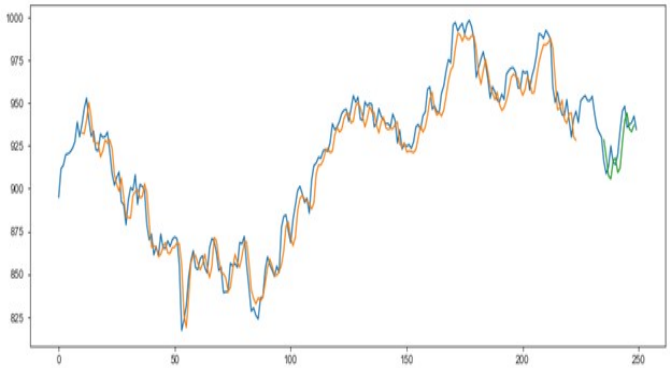

Fig. 15. CNN model - ICICI BANK data

The calculation of the Root Mean Squared Error (RMSE) score for the aforementioned models has also been conducted. It can be asserted from figure 14 that, in terms of the RMSE score, the CNN model exhibits superior performance in forecasting the Powergrid stock price.

The calculation of the Root Mean Squared Error (RMSE) score for the aforementioned models has also been undertaken. It can be concluded from figure 16 that, in terms of the RMSE score, the CNN model demonstrates superior performance in forecasting the ICICI Bank stock price.

V. Conclusion and Future Scope

In conclusion, this research paper delved into the application of various deep learning-based methods for predicting stock prices, utilizing one-year historical stock data for three distinct stocks. The algorithms considered in this study encompassed LSTM, Prophet, Convolutional Neural Network (CNN), and the LSTM-GRU Hybrid model. LSTM has traditionally been a preferred algorithm for sequential data analysis. However, this study revealed that the CNN model outperformed all other algorithms, achieving the lowest Root Mean Squared Error (RMSE) for both the Powergrid and ICICI Bank datasets. Interestingly, the LSTM-GRU Hybrid model exhibited superior performance only in the TCS dataset. These findings suggest that the CNN model may yield a more accurate and robust predictive model.

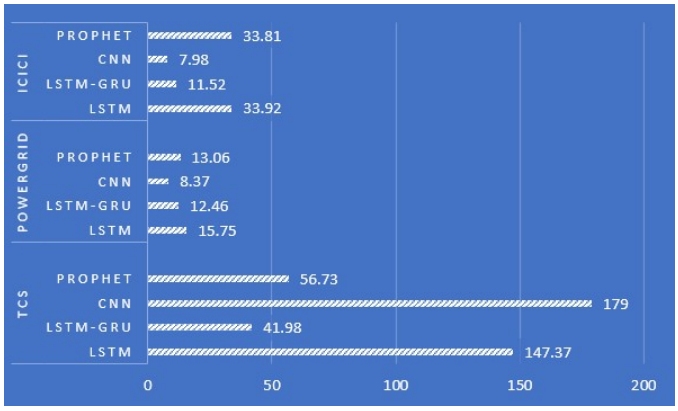

Fig. 16. RMSE Score of models - ICICI, Powergrid and TCS data

Potential avenues for future research in this domain may involve exploring the integration of Reinforcement Learning with LSTM or other hybrid approaches. Additionally, different feature selection methods could be investigated to enhance model performance. Lastly, evaluating the models on a larger dataset or in a real-time scenario could provide insights into their generalizability and practical applicability.

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